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FINAL RESEARCH REPORT — SUBMISSION COVER PAGE

PROJECT / RESEARCH TITLE

PhantomSense: Hybrid IoT and Computer Vision Framework for Detecting Human Decomposition in Flowing Water

PROGRAMME DETAILS

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SUBMITTED BY

Student Rohit Majumder

UNDER THE MENTORSHIP OF

Mentor Name Amiya Karmakar

Affiliation SalesForce

DECLARATION

We hereby declare that this report is a record of genuine work carried out by us under the guidance of the above-mentioned mentor, as part of the IEEE SMC Student Branch Chapter, KGEC — Research Internship Programme 2026, and has not been submitted elsewhere for any other purpose.

Mentor's Signature

Programme Coordinator

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1. Introduction

Detecting a human body in a water body is a difficult and time-sensitive task. In cases involving drowning, accidental deaths, suicides, or suspected murders, investigators need to locate the victim quickly for rescue, recovery, and investigation. Current search operations often rely on human divers, boats, and manual observation. These methods can take a long time and can put search teams at risk, especially when they need to cover large or deep water areas.

Several environmental conditions make detection even more difficult. Mud, algae, turbidity, and suspended particles reduce underwater visibility. Waves, ripples, reflections, and changes in lighting can distort the appearance of objects. Water movement can also carry a body away from its original location. Underwater objects and obstructions may further confuse the search because they can resemble parts of a human body. These conditions make continuous manual monitoring of large water areas difficult.

To address these challenges, we propose PhantomSense, a hybrid Internet of Things (IoT) and Computer Vision framework for detecting possible human presence in flowing water. We designed PhantomSense to combine sonar-based computer vision, IoT sensing, data processing, and real-time communication.

Our current vision component uses a YOLOv11 model to analyse sonar imagery and detect possible human presence. We also plan to add a chemical sensing layer that can detect blood and different types of rotting organic material. When the system detects a suspicious object, it can send the available information to a central server and dashboard. The system can then provide information such as the detection location, time, image evidence, and chemical details to support further investigation.

Our main goal is not to replace investigators or rescue personnel. We designed PhantomSense to provide an automated first layer of detection that can reduce manual search effort and help responsible teams investigate possible incidents more quickly.

2. Objectives

We designed PhantomSense with the following objectives:

1. We aim to continuously monitor a designated water area using imaging devices and sensors.
2. We use a YOLOv11-based computer vision model to identify possible human presence in sonar imagery.
3. We plan to develop a chemical sensing layer that can detect blood and rotting organic materials.
4. We aim to combine visual and chemical information to improve detection reliability.
5. We plan to transmit sensor and detection information to a central server through an IoT-based system.
6. We aim to provide useful information through a monitoring dashboard, including location, time, image evidence, and chemical details.
7. We plan to generate real-time alerts when the system identifies a suspicious detection.
8. We aim to preserve collected information for later forensic investigation.
9. We aim to reduce the need for continuous manual searching across large water areas.
10. We plan to use collected data to improve the system and its models over time.

3. Background and Problem Statement

Searching for a human body in flowing water creates several practical difficulties.

First, poor underwater visibility makes detection difficult. Mud, algae, turbidity, and suspended particles can hide objects and reduce the quality of visual information.

Second, water movement constantly changes the environment. Waves and ripples can distort the appearance of objects, while flowing water can move a body away from its original location. This movement increases the area that search teams need to examine.

Third, underwater objects can create misleading patterns. Rocks, branches, debris, and other obstructions may resemble parts of a human body. A system that depends on only one type of information may therefore produce incorrect results.

Finally, large water areas require considerable time and manpower for continuous manual observation. Search teams cannot practically monitor every part of a large or fast-flowing waterbody at all times.

These problems create a need for an automated system that can monitor a water environment and identify possible human-body presence. PhantomSense addresses this problem by combining computer vision and IoT-based sensing instead of depending on a single detection method.

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4. Proposed Solution: PhantomSense

We designed PhantomSense as a hybrid detection framework that combines waterbody monitoring, sonar imagery, computer vision, IoT sensors, a server, a dashboard, and an alert system.

The system continuously monitors a selected water area. Imaging devices collect information from the environment, while the IoT layer collects sensor information.

We use the computer vision component as the preliminary detection layer. The YOLOv11 model analyses sonar imagery and identifies objects that may correspond to a human body.

At the same time, we plan to use chemical sensors to collect additional information. These sensors will detect blood and different types of rotting organic material that may resemble the chemical conditions associated with a dead body.

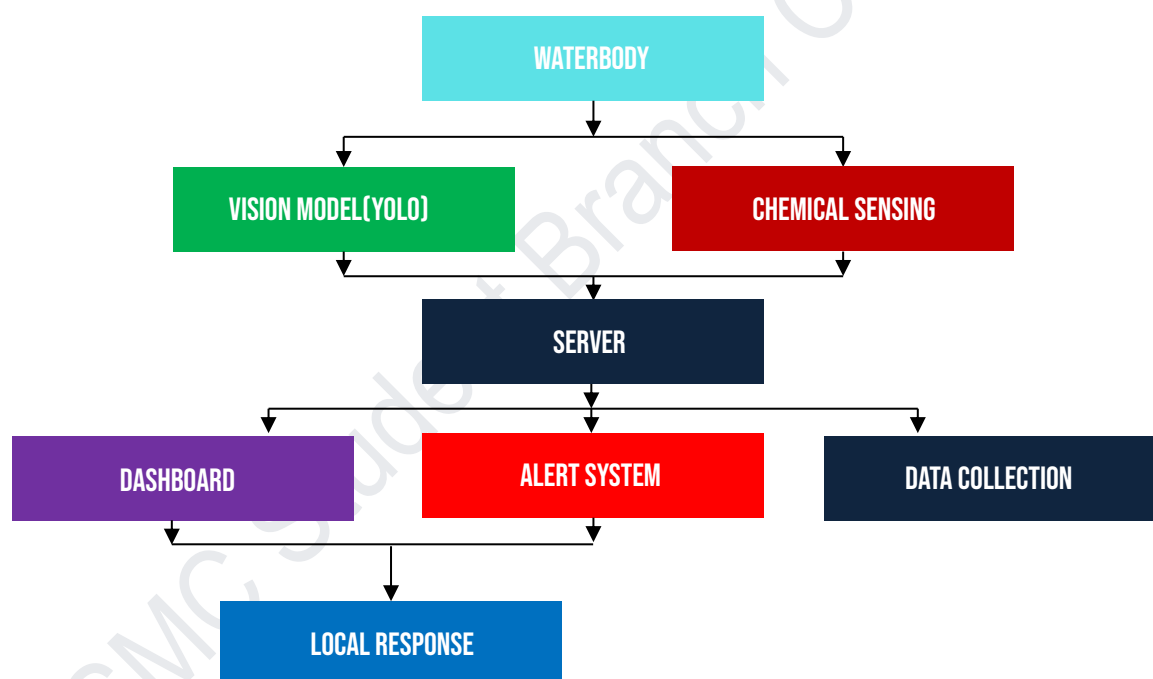
When PhantomSense detects a possible human body, the system can send the relevant information to the server. The server can analyse the available data and provide the results to the dashboard. The system can also generate an alert containing information such as the detection location and time.

5. System Architecture

We divided the PhantomSense architecture into several major components:

Waterbody → Vision Model → Chemical Sensing → Server → Dashboard → Alert System → Local Response

The architecture also includes data collection and communication between the different components.



5.1 Waterbody Monitoring

We begin the detection process by continuously monitoring the selected waterbody.

The system collects information from imaging devices, sonar scanners, and sensors. Continuous monitoring allows PhantomSense to observe the environment without requiring a person to manually inspect the water at all times.

5.2 Vision Model

We designed the vision model using YOLOv11 and use sonar images as its input.

We trained the model using the AquaScan-1K dataset. During the evaluation described in the project material, the model successfully identified human presence in a sonar scan. We also tested a horizontally inverted version of the evaluation image, and the model successfully detected the human presence there as well.

We use this model as the preliminary checking layer of the framework. When the model detects possible human presence, the system can forward the result to the server for further inspection.

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5.3 Chemical Sensing

We are currently developing the chemical sensing component of PhantomSense.

We plan to use chemical sensors to detect blood and different rotting organic materials that may resemble a dead body. We will pass the sensor readings to another classification model, which will analyse the chemical information.

We then plan to send the classification results to the server. This information can help the system determine whether the detected object may represent a human body or another source of organic material.

This component is still under development, so we may change its implementation as the project progresses.

5.4 Server and API

We designed the server as the central processing and communication component of PhantomSense.

The server will collect information through an API and analyse the available data. The system will then return a response that indicates whether it suspects a dead body or another material that may resemble a dead body, such as a dead animal or rotting meat waste.

When the system identifies a suspicious case, the server will send relevant information to the dashboard. This information can include image evidence, the location of the sighting, and chemical details.

We also plan to store the collected data. The stored information can support future model improvement and preserve evidence that forensic teams may use during an investigation.

5.5 Dashboard and Alert System

We designed the dashboard to present important detection information in one place.

When PhantomSense detects a suspicious object, the dashboard can display the image evidence, location, time, and available chemical information. The alert system can then notify the responsible personnel so that they can investigate the reported location.

This approach allows the system to move beyond simple object detection and provide information that can support a practical response.

6. Methodology

We divided the PhantomSense workflow into six main stages: continuous monitoring, data collection, computer vision analysis, IoT integration, detection and verification, and real-time alerting.

6.1 Continuous Monitoring

We continuously monitor the selected water area using cameras, sonar scanners, and sensors.

This stage allows the system to observe the environment without requiring a person to watch the water continuously.

6.2 Data Collection

We collect visual and sensor information from the water environment.

The visual pipeline provides sonar imagery for the computer vision model, while the IoT layer collects information from the available sensors.

6.3 Computer Vision Analysis

We process the collected sonar images using our YOLOv11 model.

The model searches for patterns and objects that may indicate human presence. We use the output as an initial detection rather than as final proof of a human body.

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6.4 IoT Integration

We use the IoT layer to connect the sensing components with the rest of the PhantomSense architecture. The system communicates sensor information and detection results to the monitoring infrastructure so that we can process the information centrally.

6.5 Detection and Verification

We combine the available information to improve detection reliability. For example, the vision model may identify an object that resembles a human body in sonar imagery. The chemical sensing layer can provide additional information that helps us determine whether the object may represent a human body or another source of organic material. This multi-stage process can help us reduce false alarms compared with a system that depends on only one source of information.

6.6 Real-Time Alerting

When PhantomSense identifies a sufficiently suspicious detection, the system can generate an alert. The alert can include the detection location, time, image evidence, and available chemical information. Responsible personnel can use this information to investigate the reported location.

7. Vision Model Implementation

Our current computer vision component uses YOLOv11 to detect possible human presence in sonar imagery. We trained the model using the AquaScan-1K dataset and use raw sonar images as input. During the evaluation described in the project material, the model successfully detected human presence in a sonar scan. We also tested a horizontally inverted version of the image and obtained a successful human detection. The model therefore provides the preliminary screening stage of PhantomSense. However, our project material does not provide numerical evaluation metrics such as accuracy, precision, recall, F1-score, or mean Average Precision (mAP). We therefore do not include unsupported numerical performance claims in this report. The current results show that our model can detect human presence in the evaluated sonar imagery. We need additional experiments and a larger evaluation before we can make broader performance claims.

8. Results and Discussion

Our current implementation provides a preliminary vision-based detection component for the PhantomSense framework. The YOLOv11 model successfully identified human presence in the sonar scan used during evaluation. We also tested a horizontally inverted version of the image and observed successful detection. These results show that the model can identify relevant patterns in the provided sonar imagery after a change in image orientation. We have not completed the chemical sensing layer, so we cannot report final experimental results for chemical classification. We have also not completed the full server and API implementation described in the proposed architecture. The project material states that we will design these components after we complete the chemical sensing stage. Our current project therefore combines an implemented preliminary vision component with a proposed IoT and chemical-sensing framework. The main strength of our approach comes from combining different information sources. The vision model provides one form of evidence, while chemical sensing can provide another. By combining these signals, we aim to improve the reliability of suspicious-body detection. At the same time, our current results do not allow us to claim that the complete PhantomSense framework can reliably identify dead bodies in real-world flowing rivers. We need additional testing, chemical-sensing results, and full system integration before we can make that claim.

9. Limitations and Drawbacks

The biggest limitation of PhantomSense is the possibility of false positives.

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Our system uses sonar imagery and chemical sensing, and both sources can produce misleading signals. For example, a sonar image may resemble a human body while rotting meat or another form of organic material may exist in the same area. In such a situation, the system could interpret the combined signals as evidence of a dead body and generate a false alert. This situation could create unnecessary panic and cause police or local response teams to spend resources on an incorrect detection.

Fast-flowing and oxygen-rich rivers create another challenge. According to our project material, organic evidence may fade quickly in such environments. This condition could cause the chemical sensing layer to miss important evidence and produce a false negative.

These limitations show why we should use PhantomSense as a support and early-warning system rather than as an independent system for making final forensic decisions.

Trained personnel must verify important detections before they take further action.

10. Future Scope

We plan to improve PhantomSense in several areas.

10.1 Chemical Sensing

We will complete the chemical sensing layer and integrate it with the rest of the framework. The completed layer will allow us to analyse chemical information alongside sonar-based visual information.

10.2 Server and API Development

After we complete the chemical sensing layer, we will develop the server and API system. The server will collect information from the different detection components and provide the results to the dashboard.

10.3 Drone-Based Verification

We can introduce inspection drones as an additional verification mechanism. Police or other responsible personnel could use a drone to inspect a reported location and confirm whether the system has identified an actual human body.

This approach could reduce false positives and save the time and resources that response teams might otherwise spend on unnecessary investigations.

10.4 Localized IoT Operation

We can improve the IoT infrastructure so that PhantomSense can continue working without a constant internet connection. A localized system could process and communicate detection information within the local environment instead of depending entirely on remote API requests. This improvement could make the system more practical in locations with weak or unavailable internet connectivity.

10.5 Model Improvement

We can continue collecting sonar data and use reliable detections to improve the computer vision model. Additional data can help us evaluate the model under different environmental conditions, object orientations, and water-flow conditions. More extensive evaluation can also help us understand and reduce false positives and false negatives.

11. Conclusion

We designed PhantomSense as a hybrid IoT and Computer Vision framework to help detect possible human bodies in flowing water.

We developed the project around the idea of combining different types of information instead of relying on a single detection method. Our current vision component uses YOLOv11 to analyse sonar imagery and identify possible human presence. We trained the model using the AquaScan-1K dataset, and the model successfully detected human presence in

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the sonar imagery used during the reported evaluation.

We are currently developing the chemical sensing component. We plan to use it to detect blood and other rotting organic materials and provide additional evidence for the detection process. We also plan to connect these components through a server, API, dashboard, and alert system.

Our project also identifies important limitations. Environmental conditions can create false positives and false negatives, especially in fast-flowing rivers. We therefore do not intend PhantomSense to replace human investigators. Instead, we designed it to support them by providing early detection and useful information.

In the future, we plan to complete the chemical sensing layer, develop the server and API, introduce drone-based verification, and improve the IoT system so that it can operate in more localized environments.

Ultimately, we hope PhantomSense can make difficult water searches faster and more systematic while reducing the risks and effort associated with continuous manual searching.

12. Reference:

[1] AquaScan-1K Side Scan Sonar Dataset, <https://zenodo.org/records/18771165>.

13. AI Usage Declaration

We designed the PhantomSense concept, system architecture, model architecture, and overall technical approach ourselves. We developed the project based on our own research, analysis, and implementation work. We did not use any AI tool to design the core PhantomSense architecture, model architecture, or technical concept.

We used AI-based grammar and language refinement tools only to improve the grammatical correctness, sentence structure, readability, and presentation of this report. We used these tools for grammatical refinement and language editing, not for developing the project's core ideas, architecture, model design, methodology, or technical implementation. We take responsibility for the technical content, system design, analysis, results, and claims presented in this report. We reviewed the final report to ensure that the content accurately represents our own project work.

References